Technical Statement of Work (SOW)

Technical Specification: Under Axle Weighing System (UAWS)
Manufacturer: Jackson Aircraft Weighing Systems (JAWS)

1. SYSTEM OVERVIEW

The **Jackson Aircraft Weighing Systems (UAWS)** is a turnkey, high-capacity solution engineered for the precision weighing of wide-body jet aircraft. While optimized for **Boeing 757, 767, 777, and 787** aircraft, the system's modular design ensures compatibility with a broad range of heavy-lift commercial and military aircraft.

By integrating wireless load cell technology directly with air-over-hydraulic jacking systems, the UAWS eliminates the need for traditional platform scales, significantly reducing hangar footprint setup time and increased safety to prevent back injury from lifting heavy platforms, roll on and off operations requiring heavy tow bars and tugs, preventing uncommanded rolling off and the ability to weigh in position.

2. CORE COMPONENTS

•**Wireless Sensing Array:** Five (5) high-capacity wireless load cells interrogated via a dedicated primary workstation.

•**Pneumatic-Hydraulic Jacking Set:** Five (5) specialized jacks (Models J2400-80 and J2400-100) providing 80,000 lb to 100,000 lb lift capacities respectively.

•**Advanced Data Suite:** A Windows laptop configured with **JAWS UAWS Software**, capable of generating comprehensive 8.5" x 11" weight and balance reports, CSV files and PDF out put.

•**Precision Adapters:** A suite of adapters designed to accommodate variances in axle height resulting from tire wear, inflation levels, or specific condition requirements. These are secured via a 1.75" machined mounting hole and specialized cap-screw configurations for maximum stability.

3. OPERATIONAL CHARACTERISTICS

•**Pneumatic Power:** Systems operate on standard 120 psi shop air. A three-position (Up/Neutral/Down) control valve manages the air motor, delivering smooth, incremental hydraulic lift.

•**Wireless Workflow:** The system utilizes a "Remote Wake" feature, allowing the operator to activate and zero cells from the laptop workstation once the jacks are positioned, minimizing manual interaction with the hardware.

•**Seating Verification:** The low-profile design allows for side-to-side alignment checks to ensure aircraft jack points are perfectly centered in the load cell cups before full weight is applied.

4. EFFICIENCY & PROTOCOL

The UAWS facilitates a rapid **"Lift-Level-Weigh"** cycle. The system's precision allows for quick leveling and the ability to perform "re-zero" verification lifts without moving the aircraft. In a standard maintenance environment, a full wide-body weighing procedure can be successfully completed in **under 60 minutes**.

Specifications

REQUEST FOR QUOTATION (RFQ)

Project: Under Axle Weighing System Boeing 777 (UAWS777) – For Large Jet Aircraft
RFQ Reference: NASA
Issue Date: 01/30/26

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1. SCOPE OF REQUIREMENT

This request seeks a high-capacity **Under Axle Weighing System (UAWS)** for weighing Large Jet Aircraft exceeding 300,000 lbs. The system must integrate pneumatic-over-hydraulic jacks with high-precision wireless load cells and a dedicated laptop running Windows 11.

2. COMPONENT SPECIFICATIONS

2.1 Jacking System (J2400 Series)

Five (5) stand-alone, dual-range height jacks machined for aerospace applications:

•**Inventory:** (1) J2400-80 [80K lbs] and (4) J2400-100 [100K lbs].

•**Actuation:** Self-contained air-over-hydraulic via 120 psi shop air; pneumatic solenoid directional control.

•**Portability:** Dual wheels, onboard oil tank, and adjustable handle with integrated air valve.

•**Height Profile (with 100K Cell and adapters installed):**

•**80K Jack:** Low height max of 14.625 inches

•**100K Jack:** Low height max of 15.250 inches.

2.2 Wireless Load Cells

Five (5) wireless 100,000 lb capacity cells:

•**Construction:** One-piece, multi-column machined stainless steel with "All-In-One" design (integrated PCB, antenna, and battery box).

•**Performance:** 10 lb resolution; accuracy of 1/10th of 1% or 3 divisions (whichever is greater).

•**Fitment:** Stainless steel tops for large jet points; 3/4-16 bottom thread with 1-inch mounting stud.

•**Adapters:** One set of adapters, 0, 1.0, 1.5 inches

2.3 Data Acquisition & Software (JAWS100)

•**Hardware:** Windows 11 Laptop with dedicated system USB antenna.

•**Functionality:** Log/view all active channels; custom math algorithms (summing/averaging); mapping/graphical visualization.

•**Alerting:** Visual/audible alarms for over/under range, low battery, and comm loss.

•**Reporting:** CSV format export; built-in web server for remote viewing on mobile devices.